
Automotive SoC Motor Driver

1.0 SYSTEM CONTROL UNIT (SCU)

1.1 Features

SCU supports the following features

- Clock Control
- SRAM ECC control
- IRAM ECC control
- Flash ECC control
- On-chip Diagnostics control
 - Power on Reset
 - System reset
 - Watchdog timeout
 - CPU lock up
 - GDU fault
 - PLL lock
 - PLL timeout
 - PLL source timeout
 - VD50 Undervoltage
 - VD50 undervoltage recovery
 - VD30 Undervoltage
 - VIO Undervoltage
 - SRAM ECC
 - IRAM ECC
 - Flash ECC
 - Die-overtemperature
- SCI1 control
- SCI2 control
- LIN Physical layer connection control
- Boot register
- SCU IRQ control
- Interrupt generation from on-chip fault

1.2 Introduction

The 89201 has system control unit. It contains clock control for CPU and clock gating feature for each peripheral. It allows to optimize the processing capability and power consumption.

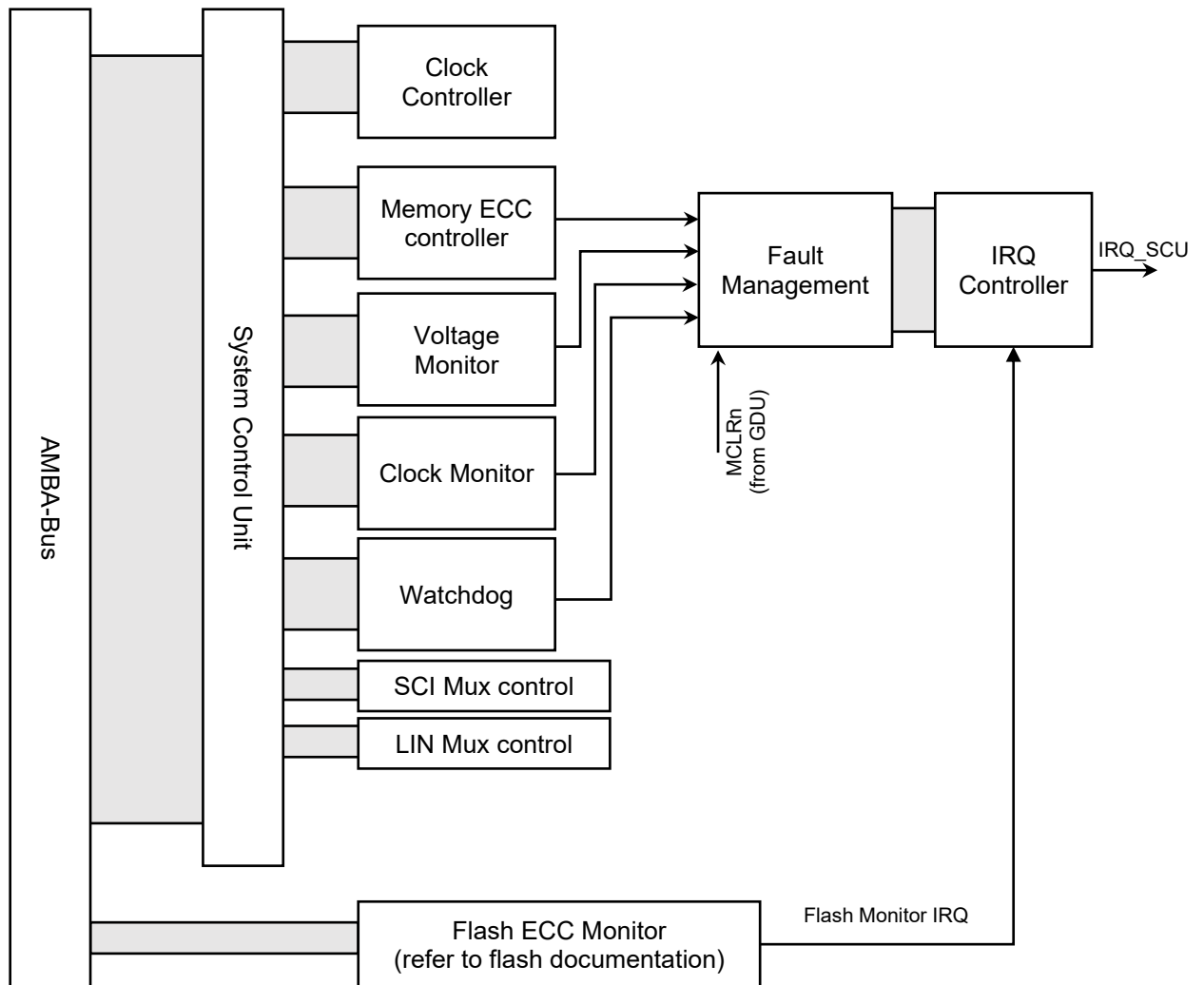
There are hardware ECC unit for each memory properties, SRAM, IRAM, Flash. The system control allows to trace failing memory location. Also, it can generate a SCU interrupt as option.

Watchdog timer allows to monitor for software execution time out check. It is also possible to generate SCU interrupt as option.

A89201 has an on-chip diagnostic for ensuring correct chip operation. When it is detected, corresponding register flag the fault. And it can generate SCU interrupt to take further action if necessary.

Also SCU includes the on-chip signal peripheral signal controls on top of each peripheral register.

2.0 BLOCK DIAGRAM



3.0 REGISTER SET

Table 1: Register map of the SCU

Base Address: 0x4000D000

Register Name	Access Type	Address offset	Reset Value
mcu_Info	R	0x0000	0x00D4 0000
ctrl	R/W	0x0008	0x0000 0000
intstat	R/W1C	0x0010	0x0000 0000
inten	R/W	0x0014	0x0000 0000
diag	R/W1C	0x0018	0x0000 0001
mcucclk	R/W	0x001C	0x0000 0000
ecc_dsa	R	0x0020	0x0000 0000
ecc_dse	R	0x0024	0x0000 0000
ecc_isa	R	0x0028	0x0000 0000
ecc_ise	R	0x002C	0x0000 0000
wdt	W	0x0030	0x0000 0000
wdt_period	R/W	0x0034	0x0000 0000
sel_uart0	R/W	0x0038	0x0000 0000
sel_uart1	R/W	0x003C	0x0000 0000
sel_spi0	R/W	0x0040	0x0000 0000
sel_spi1	R/W	0x0044	0x0000 0000
sel_lin	R/W	0x0048	0x0000 0000
sel_lctx	R/W	0x0054	0x0000 0000
debug	R/W	0x0058	0x0000 0000

3.1 Register Definition

3.1.1.2 MCU Info

Offset Address = 0x0000

Reset Value = 0x00D4_0000

Bit31	Bit30	Bit29	Bit28	Bit27	Bit26	Bit25	Bit24
Res	Res	Res	Res	Res	Res	Res	Res
R	R	R	R	R	R	R	R

Bit23	Bit22	Bit21	Bit20	Bit19	Bit18	bit17	Bit16
Res	Res	Res	Res	revnum			

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	Res	Res	Res	Res	Res	Res
R	R	R	R	R	R	R	R

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Res	Res	Res	Res	Res	Res	Res	Res
R	R	R	R	R	R	R	R

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
revnum	19:16	MCU mask revision number

3.1.1.2 CTRL Register

Offset Address = 0x0008

Reset Value = 0x0000_0000

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
			viomon	nohlt	plloff	ida	mclrn_en
U	U	U	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
mclrrst	sci1mode	sci0mode	Res	rsten	wdten	lken	dda
RW	RW	RW	RW	RW	RW	R/W	R/W

ev

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
dda	0	Data direct access mode enable allows the user to access the ECC data of the SRAMs. 0: Disabled 1: Enabled
lken	1	Enable a CPU Lockup to cause a warm reset 0: Disabled 1: Enabled
wdten	2	Enable Watchdog Timer 0: Disabled 1: Enabled Note: After BootROM execution, wdten is set to 1 and watchdog timer is enabled.
rsten	3	Watchdog Timer reset Enable (wdten shall be set) 0: Reset disabled 1: Reset enabled Note1: When this is 0, interrupt is generated. If it is 1, then hardware reset will be generated. When the reset is asserted, the ongoing software execution will be suspended and start from the reset condition. Note2: After BootROM execution, rsten is set to 1 and upon watchdog timeout, the MCU will reset if the user application code doesn't handle watchdog.
Res	4	Read as 0, do not write 1 to this bit.
sci0mode	5	Select Serial communication for SCU1 0: SPI mode 1: UART mode Note: When one of them is selected, the other mode cannot be used. Please refer application section for more information.
sci1mode	6	Select Serial communication for SCU2 0: SPI mode 1: UART mode Note: When one of them is selected, the other mode cannot be used. Please refer application section for more information.
mclrrst	7	Warm-reset enable from MCLRn 0: Generate NMI interrupt 1: Generate Warm-reset Note: MCLRn is inter-connected bonding connection from GDU. Please refer GDU section for further detail. See the diagram below for the diagram.
mclrn_en	8	Enable MCLRn Enable to trigger the warm-reset or NMI interrupt insertion upon mclrn terminal assertion 0: Disabled 1: Enabled
ida	9	Instruction RAM ECC direct access enable 0: Disabled 1: Enabled
plloff	10	PLL Disable 0: Enabled 1: Disabled When the PLL is turned off, clock source will be changed to clk_wd and power consumption will be reduced. Before turning off PLL, make sure to set clock to clk_wd by adjusting selwd. Allow 4 clock cycles delay before turning off PLL after change of clock to clk_wd.

nohlt	11	SCI FIFO active when debug HALTED is set 0: FIFO frozen 1: FIFO Active
viomon	12	VIO undervoltage monitor enable 0: Disabled 1: Enabled

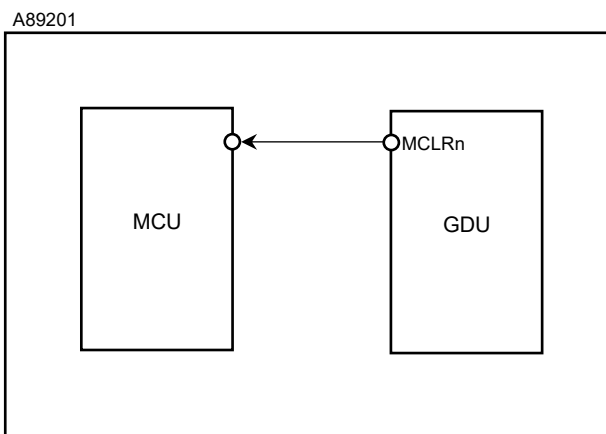


Fig3-1-1-2-1: MCLRn connection detail

3.1.1.2 INTSTAT Register

Offset Address = 0x0010

Reset Value = 0x0000

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
					vd50uvr	lostlock	wdt
U	U	U	U	U	R1C	R1C	R1C

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
isdcd	issect	dsded	dssec	ovtemp	viouv	vd33uv	vd50uv
R1C	R1C	R1C	R1C	R1C	R1C	R1C	R1C

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
vd50uv	0	VD50 terminal undervoltage 0: Not detected 1: Detected Note: Please refer the datasheet for the threshold described as $V_{VD50_UV_fal}$
vd33uv	1	VD33 terminal undervoltage 0: Not detected 1: Detected Note: Please refer the datasheet for the threshold described as $V_{VD33_UV_fal}$
viouv	2	VDIO terminal undervoltage 0: Not detected 1: Detected Note: Please refer the datasheet for the threshold described as $V_{VIO_UV_fal}$
ovtemp	3	Over temperature 0: Not detected 1: Detected Note: Please refer the datasheet for the threshold described as T_{OT}
dssec	4	Data SRAM ECC Single Error

		0: Not detected 1: Detected Note: Please refer the datasheet for the threshold described as V_{VD33_UV}
dsded	5	Data SRAM ECC Double Error 0: Not detected 1: Detected
issec	6	Instruction SRAM ECC Single Error 0: Not detected 1: Detected
isded	7	Instruction SRAM ECC Double Error 0: Not detected 1: Detected
wdt	8	Watchdog Timer Overflow 0: Not detected 1: Detected
lostlock	9	PLL loss of lock 0: Not detected 1: Detected
vd50uvr	10	VD50 undervoltage recovery 0: Not detected 1: Detected Note: when VD50 rises above $VD_{50_UV_Ris}$(check datasheet), this bit will be set to indicate the 5V supply is functional

Note: The electrical parameters are described in the datasheet.

3.1.1.2 INTEN Register

Offset Address = 0x0014
Reset Value = 0x0000

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
					vd50uvr	lostlock	wdt
U	U	U	U	U	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
isded	issec	dsded	dssec	ovtemp	vdiov	vd33uv	vd50uv
RW	RW	RW	RW	RW	RW	RW	RW

Legend

R: Read
W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0
Res: Reserved bit

Field	Bits	Description
vd50uv	0	VD50 terminal undervoltage 0: Disabled 1: Enabled
vd33uv	1	VD33 terminal undervoltage 0: Disabled 1: Enabled
vdiov	2	VDIO terminal undervoltage 0: Disabled 1: Enabled
ovtemp	3	Over temperature 0: Disabled 1: Enabled
dssec	4	Data SRAM ECC Single Error 0: Disabled 1: Enabled
dsded	5	Data SRAM ECC Double Error 0: Disabled 1: Enabled
issec	6	Instruction SRAM ECC Single Error 0: Disabled 1: Enabled
isded	7	Instruction SRAM ECC Double Error

		0: Disabled 1: Enabled
wdt	8	Watchdog Timer Overflow 0: Disabled 1: Enabled
lostlock	9	PLL loss of lock 0: Disabled 1: Enabled
vd50uvr	10	VD50 undervoltage recovery 0: Disabled 1: Enabled

Note: Detailed block diagram in the application section.

3.1.1.2 DIAG Register

Offset Address = 0x0018

Reset Value = 0x0001

The DIAG register reports diagnostics and the cause of the last warm reset. The warm reset bits are cleared by writing one.

Note that the warm resets do not generate interrupts since they generate a reset to the ARM and the AMBA system instead. A warm reset does not restart of the boot sequence or reset the test interface.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
		srcto	pllto	pllok			
		R1C	R1C	R	Res	Res	Res

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
		pcfg	mclr	lkup	wdt	sys	por
Res	Res	R	R1C	R1C	R1C	R1C	R1C

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
por	0	Power On reset 0: Not detected 1: Detected
sys	1	A system request was the cause of the last warm reset 0: Not detected 1: Detected
wdt	2	The WDT was the cause of the last warm reset 0: Not detected 1: Detected
lkup	3	A CPU LOCKUP was the cause of the last warm reset 0: Not detected 1: Detected
mclr	4	MCLR from the GDU generates an NMI or warm reset 0: Not detected 1: Detected
pcfg	5	Powerup pin configuration 0: TEST tied to GND: Function Mode 1: TEST tied to 3.3V: Function Mode with SWD access
pllok	11	PLL locked and PLL source OK 0: Not detected 1: Detected
pllto	12	PLL timeout detection 0: Not detected 1: Detected Note: Please refer the datasheet for the threshold
srcto	13	PLL source timeout 0: Not detected 1: Detected Note: Please refer the datasheet for the threshold

3.1.1.2 MCUCCLK Register

Offset Address = 0x001C
Reset Value = 0x0000

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
	sci1	sci0	gtu	gpio	dau	gdu	cau
U	RW	RW	RW	RW	RW	RW	RW

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
dma	-	amct	pwm	selwd	freq[2:0]		
RW	RW	RW	RW	RW	RW	RW	RW

Legend

R: Read

W: Write

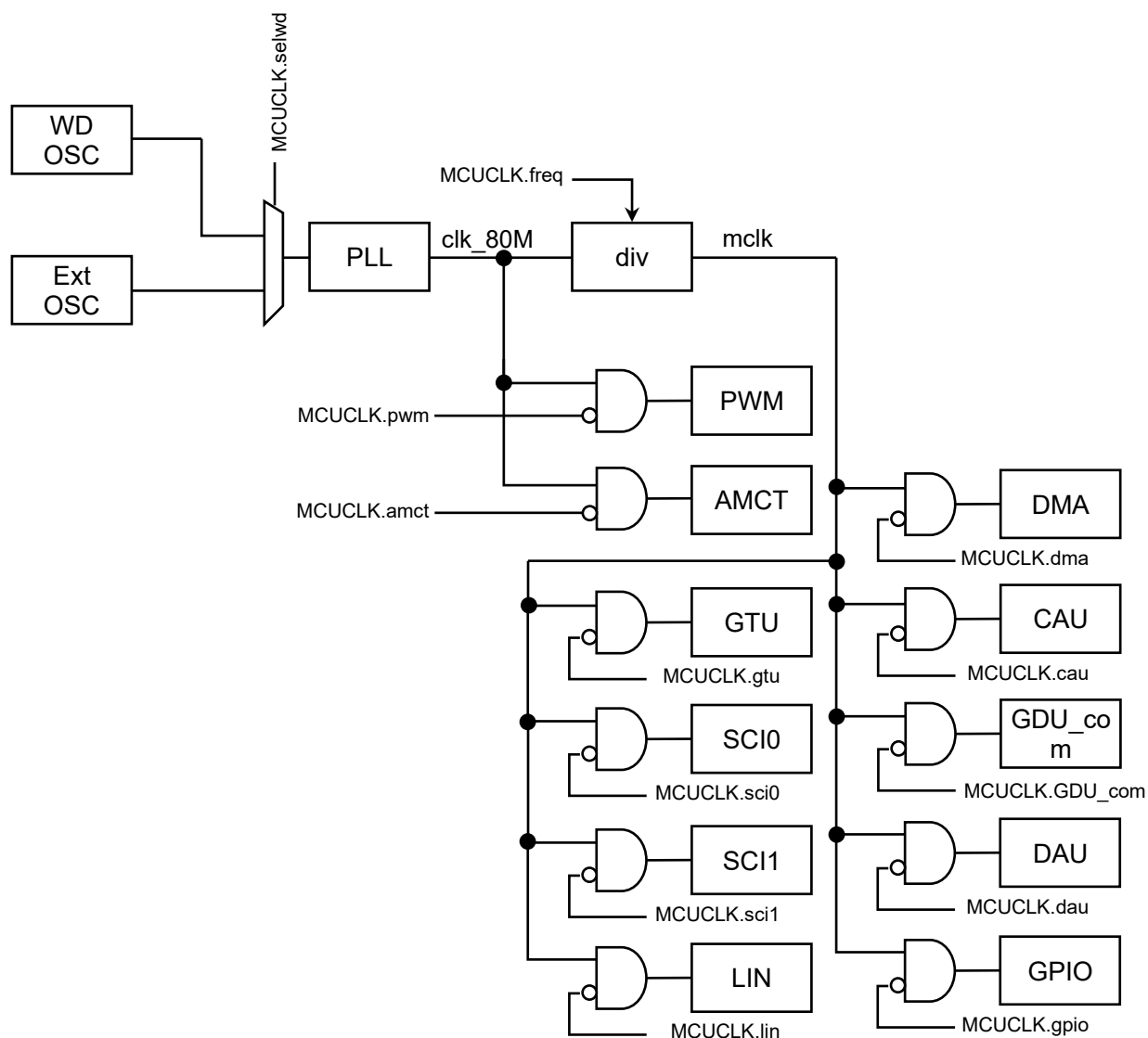
1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
freq	2:0	Divide clk_pll by value shown below to generate mclk. The mclk is then used to generate hclk. 000: 1 (40MHz) 001: 2 (20MHz) 010: 3 (10MHz) 011: 8 (5MHz) 100: 16 (2.5MHz) 101: 32 (1.25MHz) 110: 64 (625kHz) 111: 128 (322kHz)
selwd	3	Select CLK_WD as MCLK. This should be done before turning PLL off.
pwm	4	PWM clock disable 0: Enabled 1: Disabled
amct	5	AMCT clock disable 0: Enabled 1: Disabled
dma	7	DMA clock disable 0: Enabled 1: Disabled
cau	8	CAU clock disable 0: Enabled 1: Disabled
gdu	9	GDU serial communication clock disable 0: Enabled 1: Disabled
dau	10	DAU clock disable 0: Enabled 1: Disabled
gpio	11	GPIO clock disable 0: Enabled 1: Disabled
gtu	12	GTU clock disable 0: Enabled 1: Disabled
sci0	13	SCI0 clock disable 0: Enabled 1: Disabled
sci1	14	SCI1 clock disable 0: Enabled 1: Disabled
lin	15	LIN clock disable 0: Enabled 1: Disabled

Note: Clock tree is listed in the figure below:



3.1.1.2 ECC_DSA Register

Offset Address = 0x0020
Reset Value = 0x0000_0000

The ECC_DSA register reports the Data SRAM Address where the ECC error was detected.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
addr[14:8]							
U	R	R	R	R	R	R	R

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
addr[7:0]							
R	R	R	R	R	R	R	R

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
addr <14:0> ⁽¹⁾	14:0	Data SRAM ECC Error Address

3.1.1.2 ECC_DSE Register**Offset Address** = 0x0024**Reset Value** = 0x0000_0000

The ECC_DSE register reports the Data SRAM ECC syndrome

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
ecc[6:0]							
	R	R	R	R	R	R	R

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
ecc ⁽²⁾	6:0	Data SRAM ECC Syndrome

3.1.1.2 ECC_ISA Register**Offset Address** = 0x0028**Reset Value** = 0x0000_0000

The ECC_SSA register reports the Instruction SRAM Address where the ECC error was detected.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
addr[15:8]							
U	R	R	R	R	R	R	R

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
addr[7:0]							
R	R	R	R	R	R	R	R

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
addr <14:0> ⁽¹⁾	15:0	Instruction SRAM ECC Error Address

3.1.1.2 ECC_ISE Register**Offset Address** = 0x002C**Reset Value** = 0x0000_0000

The ECC_ISE register reports the Instruction SRAM ECC syndrome

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	ecc[6:0]						
U	R	R	R	R	R	R	R

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
ecc ¹	6:0	Instruction SRAM ECC Syndrome

3.1.1.2 WDT Register

Offset Address = 0x0030

Reset Value = 0x0000_0000

The WDT register is used to restart the WDT countdown register.

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	-	-	-	rstrt
U	U	U	U	U	U	U	W

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
rstrt	0	Watchdog restart, watchdog counter is reloaded

3.1.1.2 WDT_PRD Register

Offset Address = 0x0034

Reset Value = 0x0000_0000

The WDT_PERIOD register value is loaded into watchdog counter whenever 1 is written to the WDT_CTRL.RSTRT flag. The watchdog counts down in approximately 1ms steps (assuming that MCLK is 40MHz) and generates a timeout on reaching zero.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	cnt[10:8]		
U	U	U	U	U	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
cnt[7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
cnt	10:0	Watchdog countdown value in step of 1ms. Count time (ms) = cnt + 1 Note: After bootROM execution, the cnt is set to 0x7FF.

3.1.1.2 SEL_UART0 Register

Offset Address = 0x0038

Reset Value = 0x0000_0000

The SEL_UART0 register allows the user to select to the RX input to UART0.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	rx[3:0]			
U	U	U	U	R/W	R/W	R/W	R/W

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
rx <3:0>	3:0	Select GPIO[15:0] as the UART0 rx input. 0: D0 1: D1 2: D2 3: D3 4: D4 5: D5 6: D6 7: D7 8: D8 (LIN-Rx) 9: D9 (LIN-Tx) 10: D10 (HA) 11: D11 (LA) 12: D12 (HB) 13: D13 (LB) 14: D14 (HC) 15: D15 (LC) <i>Note: GPIO need to be configured as digital input. Please refer GPIO section. Please also refer the SCI section for more signal integrity.</i>

3.1.1.2 SEL_UART1 Register

Offset Address = 0x003C

Reset Value = 0x0000_0000

The SEL_UART1 register allows the user to select to the RX input to UART1.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	rx[3:0]			
U	U	U	U	R/W	R/W	R/W	R/W

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
rx <3:0>	3:0	Select GPIO[15:0] as the UART1 rx input. 0: D0 1: D1 8: D8 (LIN-Rx) 9: D9 (LIN-Tx)

2: D2	10: D10 (HA)
3: D3	11: D11 (LA)
4: D4	12: D12 (HB)
5: D5	13: D13 (LB)
6: D6	14: D14 (HC)
7: D7	15: D15 (LC)

Note: GPIO need to be configured as digital input.
Please refer GPIO section. Please also refer the SCI
section for more signal integrity.

Offset Address = 0x0040
Reset Value = 0x0000_0000

3.1.1.2 SEL_SPI0 Register

The SEL_SPI0 register allows the user to select to the RX input to SPI0.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	csin[3:0]			
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
sckin[3:0]				rx[3:0]			
				R/W	R/W	R/W	R/W

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
rx <3:0>	3:0	Select GPI[15:0] as the SPI0 rx input. 0: D0 1: D1 2: D2 3: D3 4: D4 5: D5 6: D6 7: D7 Note: GPIO need to be configured as digital input. Please refer GPIO section. Please also refer the SCI section for more signal integrity.
sckin <3:0>	7:4	Select GPI[15:0] as the SPI0 sckin input. 0: D0 1: D1 2: D2 3: D3 4: D4 5: D5 6: D6 7: D7 Note: GPIO need to be configured as digital input. Please refer GPIO section. Please also refer the SCI section for more signal integrity.
csin <3:0>	11:8	Select GPI[15:0] as the SPI0 csin input. 0: D0 1: D1 2: D2 3: D3 4: D4 5: D5 6: D6 7: D7 Note: GPIO need to be configured as digital input. Please refer GPIO section. Please also refer the SCI section for more signal integrity.

3.1.1.2 SEL_SPI1 Register

Offset Address = 0x0044

Reset Value = 0x0000_0000

The SEL_SPI1 register allows the user to select to the RX input to SPI1.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	csin[3:0]			
U	U	U	U	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
sckin[3:0]				rx[3:0]			
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
rx <3:0>	3:0	Select GPI[15:0] as the SPI1 rx input. 0: D0 1: D1 2: D2 3: D3 4: D4 5: D5 6: D6 7: D7 Note: GPIO need to be configured as digital input. Please refer GPIO section.
		8: D8 (LIN-Rx) 9: D9 (LIN-Tx) 10: D10 (HA) 11: D11 (LA) 12: D12 (HB) 13: D13 (LB) 14: D14 (HC) 15: D15 (LC)
sckin <3:0>	7:4	Select GPI[15:0] as the SPI1 sckin input. 0: D0 1: D1 2: D2 3: D3 4: D4 5: D5 6: D6 7: D7 Note: GPIO need to be configured as digital input. Please refer GPIO section.
		8: D8 (LIN-Rx) 9: D9 (LIN-Tx) 10: D10 (HA) 11: D11 (LA) 12: D12 (HB) 13: D13 (LB) 14: D14 (HC) 15: D15 (LC)
csin <3:0>	11:8	Select GPI[15:0] as the SPI1 csin input. 0: D0 1: D1 2: D2 3: D3 4: D4 5: D5 6: D6 7: D7 Note: GPIO need to be configured as digital input. Please refer GPIO section.
		8: D8 (LIN-Rx) 9: D9 (LIN-Tx) 10: D10 (HA) 11: D11 (LA) 12: D12 (HB) 13: D13 (LB) 14: D14 (HC) 15: D15 (LC)

3.1.1.2 SEL_LIN Register

Offset Address = 0x0048

Reset Value = 0x0000_0000

The SEL_LIN register allows the user to select to the RX input to the LIN.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-------	-------	-------	-------	-------	-------	------	------

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
				gpi[3:0]			
				RW	RW	RW	RW

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
gpi	3:0	Select GPI[15:0] as the LIN rx input. LCRX is hardwired to GPIO8 on the die-to-die interface and LCTX is hardwired to GPIO9.

3.1.1.2 SEL_LCTX Register

Offset Address = 0x0054

Reset Value = 0x0000_0000

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
						dout	direct
U	U	U	U	U	U	R/W	R/W

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
direct	0	0: LCTX output defined by LIN data-link layer 1: LCTX output is forced by.dout value below
dout	1	Data value used when SEL_LCTX.DIRECT is set.

3.1.1.2 Debug Register

Offset Address = 0x0058

Reset Value = 0x0000_0000

The DEBUG register can be used to disable the SWD.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
							dis
U	U	U	U	U	U	U	R/W

Legend

R: Read

W: Write

1C: Write 1 to clear

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
dis	0	SWD port Disable 0: SWD is enabled 1: SWD is disabled Note: If this port is disabled by user code, SWD port might be permanently disabled.

4.0 APPLICATION INFORMATION

4.1 Watchdog Feature



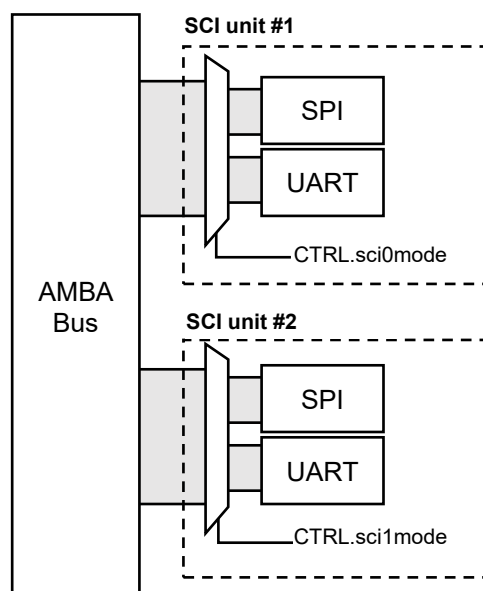
The Watchdog Timer can be used to prevent software becoming deadlocked or trapped in a loop. In normal operation the user reloads the watchdog at regular intervals before the watchdog timeout occurs. If a timeout does occur, the watchdog timer can either generate an interrupt or a reset (these two modes are exclusive).

The watchdog function must be enabled by setting 1 on the CTRL.wdten before it can countdown and potentially generate watchdog interrupts and resets. The watchdog timer is reset to the (WDT_PRD.cnt + 1) ms value when 1 is written to WDT.rstrt. For example, when counter is set to 10, the counter time out period is set to 11ms. If WDT.rstr is not set to 1 within the timeout period described above, then the watchdog time-out will be generated. If the CTRL.rsten is enabled, then a watchdog timeout generates a reset. When Reset occurs, all registers are returned to their default value. If CTRL.rsten is not enabled, and interrupt condition for watchdog is enabled, then the interrupt occurs. Before exiting the interrupt loop the watchdog must be restarted by writing 1 to WDT.rstr or otherwise the interrupt condition is permanently latched.

If CTRL.wdten is not enabled, then watchdog never generates interrupts or resets and the countdown value remains at the reset value.

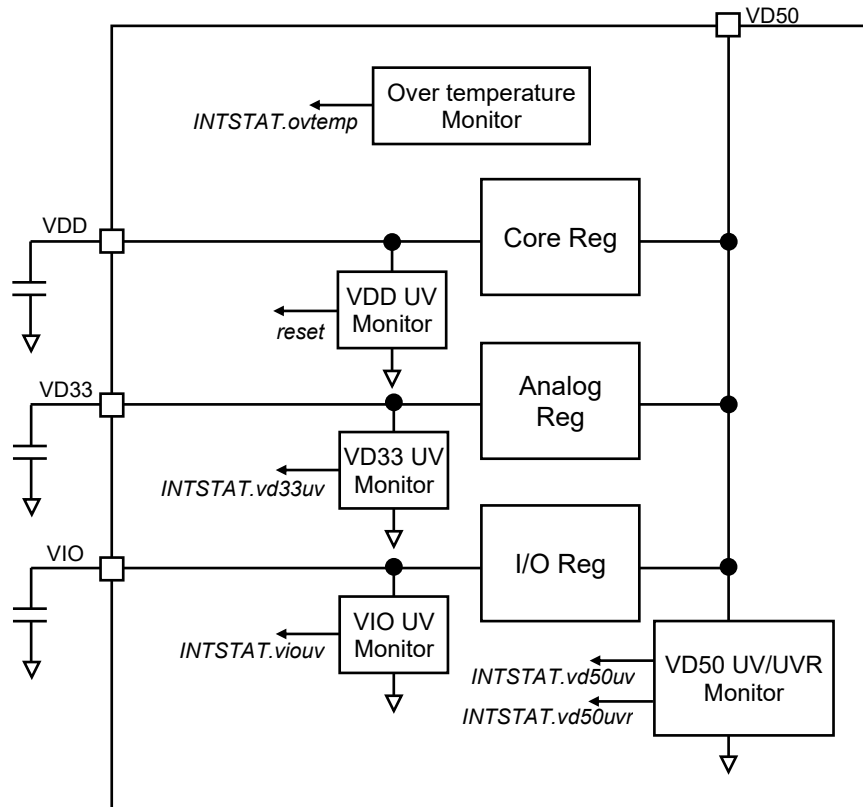
Note: If the CPU is on halt due to SWDP. The counter will be frozen until CPU restart.

4.2 SCI Mode



The SCI unit contains SPI and UART feature and feature can be chosen by CTRL.sciXmode. If one of feature is selected, the other function cannot be used at the same time. For each function detail, please refer SCI section.

4.3 Power supply monitor block diagram



MCU has single power supply input VD50 terminal. Also there are internal regulator for VDD, VD33 and VIO.

VD50 is the main power supply for MCU. The device take a power supply from VD50 to generate necessary regulated power supply for the correct operation. When the VD50 is recovered from undervoltage `vd50uvr` is set in the status register and when the undervoltage condition occurs `vd50uv` is set in the status register.

VDD is used for core-logic supply. This power supply is necessary to operate the CPU and necessary logic properly. When the VDD is in undervoltage condition, the device is in reset condition to make sure device provide correct operation.

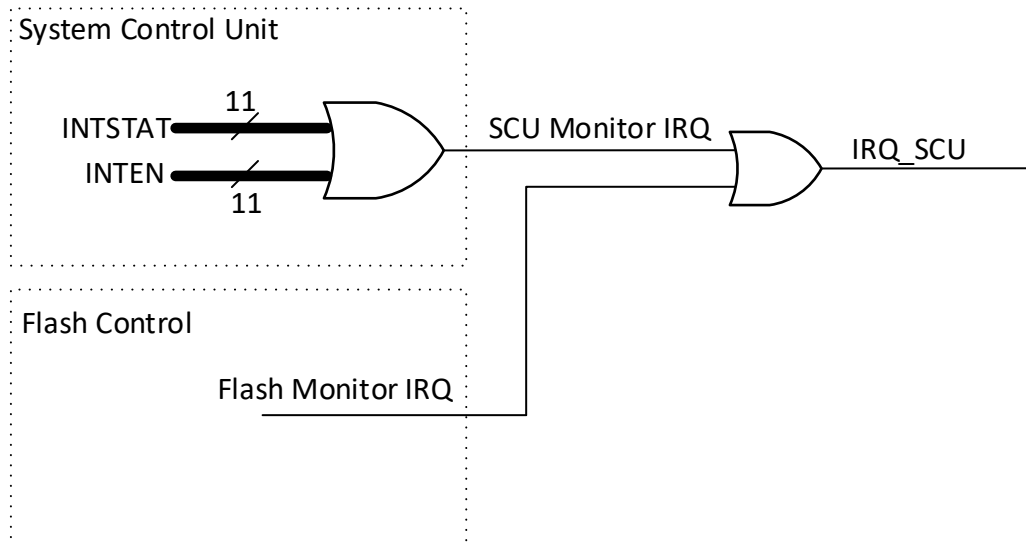
VD33 is used internal analog related items, such like ADC, clock, analog reference related items. When the VD33 is in undervoltage, the VD33 monitor is capture the event and report to system control unit. The device may operate a corrective action.

VIO is used to supply the GPIO power supply. If the VIO undervoltage condition is present, the `INTSTAT.viouuv` will be flagged.

A89201 has internal temperature sensor to monitor the die-temperature. When the overtemperature is detected, `INTSTAT.ovtemp` will be flagged.

4.4 Interrupt Management

The interrupt management for system control unit is shown in the figure below. When NVIC for system control unit is enabled, the status information from either system control peripheral or flash control peripheral can generate interrupt if the interrupt enable bit for a desired status information is set. The status information from either of the module must be checked when interrupt is generated to find the source of the interrupt condition.



Revision Control:

July 21/2023	Initial Rev
July 31/2024	Added more description for steps need to be taken before turning off PLL. Corrected the offset value for MCUCCLK register Updated description for VD50_OV
Sep 23/2024	Changed VD50OV to VD50uvr Added flash register map Added flash control register Added flash status and interrupt registers
Nov 19/2024	Removed reference to CAN for this variant of product Updated description for SEL_LCTX Register Added flash monitor in the top block diagram for fault management Corrected Debug register offset value
Feb 19/2025	Corrected bit positions for SEL_SPI0 and SEL_SPI1 tables.
July 05/2025	Updated Top level block diagram Updated CTRL register Added count time calculation for watchdog period counter Added more description to watchdog section Removed flash control and flash status information and moved them to flash control documentation (FLI) Added interrupt management section
Feb 25/2026	Added MCU Info register Added note to CTRL register to indicate the change for watchdog parameters after bootROM execution. Added note for WDT PRD Register value after bootROM execution.

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